ICPR 2026 Competition on VISual Tracking in Adverse Conditions

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Abstract. Object tracking in real-world environments is challenging due to adverse weather conditions such as rain and haze. These conditions reduce visibility, introduce noise, and degrade image quality, making tracking less accurate. In this project, we aim to build a robust object tracking system that can handle such conditions effectively. We use a combination of preprocessing techniques (diffusion-inspired) and a tracking algorithm (SiamFC-style tracker) to improve tracking performance. The performance of the system is evaluated using the Quality Precision (QP) metric.

1 Main Idea and Overall Pipeline

The proposed system is designed to perform robust object tracking under challenging weather conditions such as rain and haze. The overall pipeline consists of sequential stages including input frame processing, image enhancement, object tracking, bounding box prediction, and evaluation. To improve visibility and tracking accuracy, diffusion-inspired preprocessing techniques are applied before feeding the frames into the tracker.

The end-to-end pipeline consists of the following steps:

1. Data Preprocessing: Each input frame is enhanced using a combination of diffusion-inspired techniques. Non-local Means Denoising is applied to reduce rain noise, Bilateral Filtering is used to smooth the image while preserving important edges, and CLAHE (Contrast Limited Adaptive Histogram Equalization) is employed to improve contrast in hazy conditions. These steps collectively enhance object visibility and feature clarity.

2. Tracking (SiamFC-style): A lightweight tracking approach inspired by SiamFC is used. The tracker is initialized using the bounding box from the first frame, from which an object template is extracted. In subsequent frames, a local search region is defined, and template matching is performed using correlation

(matchTemplate) to locate the most similar region. The position of the object is updated iteratively for each frame.

3. Bounding Box Prediction: For every frame, the tracker outputs a predicted bounding box in the format [x, y, w, h], representing the object location. These predictions are generated sequentially and used for both evaluation and final submission.

4. Evaluation Metric (QP): The performance of the tracker is evaluated using the Quality Precision (QP) metric. It measures the distance between the predicted object center and the ground truth center. If the distance is within a predefined threshold, the prediction is considered correct. A QP score of 1 indicates perfect tracking performance, while a score closer to 0 reflects poor tracking accuracy.

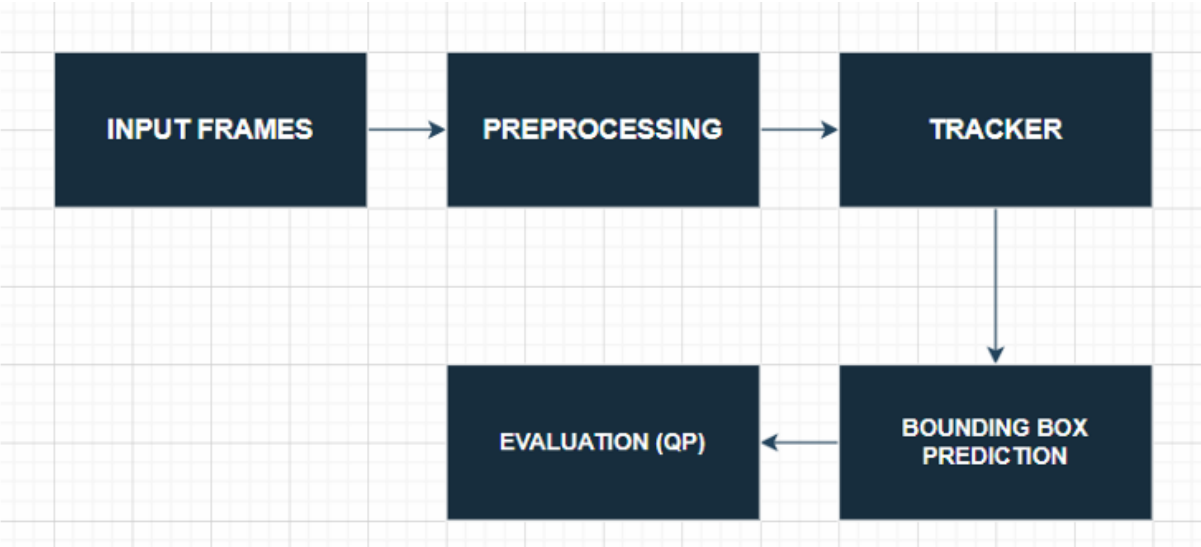
2.Training / Implementation Details

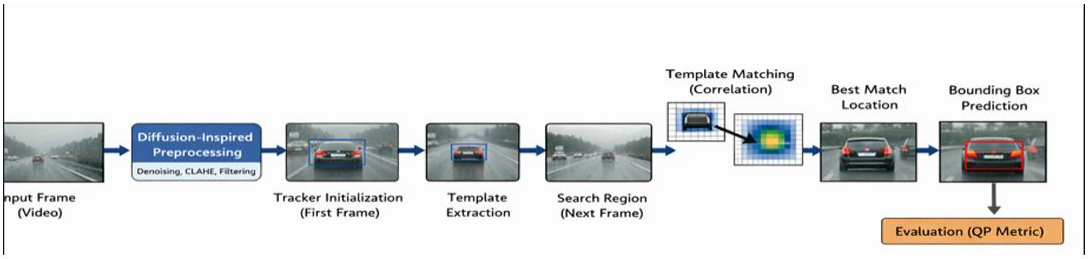
The proposed system does not involve a traditional training phase with epochs or optimization. Instead, a SiamFC-style tracking approach is used, where the tracker operates directly on video frames.

The first frame is used to initialize the tracker using the ground truth bounding box. A template of the target object is extracted and used for tracking in subsequent frames. For each new frame, a local search region is defined around the previous position, and template matching is performed using a correlation-based method.

The location with the highest similarity score is selected as the new object position, and the process is repeated for all frames. Since this is a lightweight tracking approach, no batch training, loss functions, or parameter optimization are required.

3.MODEL ARCHITECTURE





"We designed a SiamFC-style tracking pipeline where similarity between template and search region is used for tracking. Instead of a deep CNN, we used a lightweight template matching approach."

4. External Data and Pre-trained Models

We did not use any external datasets for training the person re-identification specific task; only the official TVRID training partition was used. However, to leverage transfer learning and prevent overfit ting, the osnet_x1_0 backbone was initialized with ImageNet pre-trained weights. Transforming the depth data into 3-channel JET pseudo-color images aligns the input distribution closer to the visual domain, allowing the ImageNet pre-trained filters to successfully detect edges and shapes from the colorized depth gradients.

5. Code and Reproducibility

The code and instructions to reproduce the results can be found at:
https://github.com/gompavasavi/vistac_-2_extreme_track